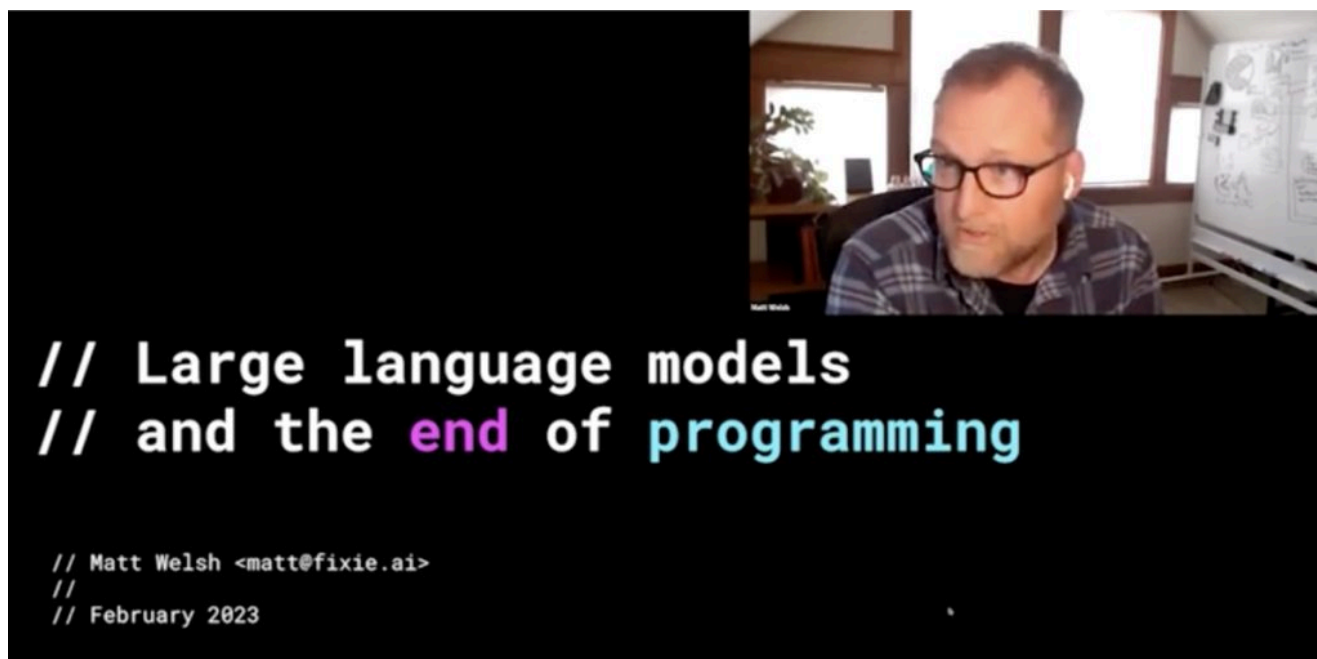


РАЗРАБОТКА ПРОГРАММНОГО ОБЕСПЕЧЕНИЯ

Программирование все равно отстой — Мэтт Уэлш о конце эпохи программирования

Бывший профессор информатики и руководитель инженерного отдела Google считает, что генеративный искусственный интеллект положит конец программированию в течение трёх лет.

21 февраля 2023 года, 10:04 [Ричард Макманус](#)



В этом месяце **Мэтт Уэлш**, бывший профессор информатики в Гарварде, выступил на виртуальной встрече **Чикагской ассоциации вычислительной техники** (ACM) и рассказал о своем тезисе, согласно которому ChatGPT и GitHub Copilot знаменуют начало **конца программирования**.

По мнению Уэлша, программирование вот-вот превратится из профессии, которой занимаются люди, в профессию, которой будут заниматься роботы, благодаря таким технологиям, как ChatGPT и Copilot. По его мнению, программистам придется стать «учителями» для программ с искусственным интеллектом — или, возможно, менеджерами по продукту или ревьюерами кода. Эти две роли, по его мнению, относительно безопасны для людей. Но не

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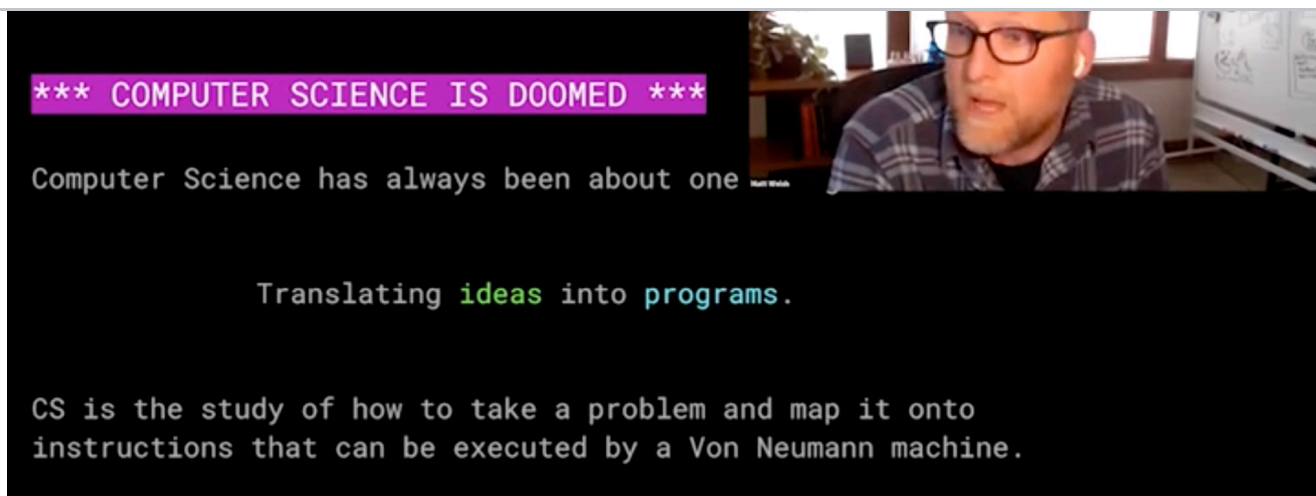
Технологии развиваются стремительно, не пропустите ни одного выпуска. Подписывайтесь на наш канал на YouTube, чтобы смотреть все наши подкасты, интервью, демонстрации и многое другое.

Подписка

Уэлш, занимавший руководящие инженерные должности в Google и Apple, написал **статью на эту тему** для журнала Communications of the ACM за январь 2023 года. Однако до этой видеопрезентации в ACM не было известно, что Уэлш основал стартап, чтобы доказать свою теорию. **Fixie.ai**, генеральным директором которой он является, позиционирует себя как «платформа автоматизации для больших языковых моделей». Она призвана помочь компаниям разрабатывать собственное программное обеспечение с использованием, как вы уже догадались, ChatGPT и аналогичных программ машинного обучения.

«Информатика обречена»

В начале своей презентации Уэлш заявил, что «информатика обречена».



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«Я думаю, что эта сфера радикально изменится, — объяснил он. — Если задуматься [о том], что такое информатика как дисциплина? Она всегда была сосредоточена на одном главном — воплощении идей в программах».

Далее он сказал, что за последние 60 с лишним лет компьютерное программирование развивалось в направлении упрощения понимания кода для людей. Он сравнил Фортран 1957 года с Бейсиком 1964 года, а затем с языком Rust 2010 года, но отметил, что с точки зрения удобства использования для программистов-людей особой разницы между ними нет.

Rust (2010)

```
fn neighbour_counts(col: &Colony) -> HashMap<Cell, i32> {  
    let mut ncnts = HashMap::new();  
    for cell in col.iter().flat_map(neighbours) {  
        *ncnts.entry(cell).or_insert(0) += 1;  
    }  
    ncnts  
}  
  
fn generation(col: Colony) -> Colony {  
    neighbour_counts(&col)  
        .into_iter()  
        .filter_map(|(cell, cnt)|  
            match (cnt, col.contains(&cell)) {  
                (2, true) |  
                (3, ..) => Some(cell),  
                _ => None  
            })  
        .collect()  
}
```



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“Programs that people are developing today are just as complex, just as hard to maintain, just as difficult to understand and just as full of bugs as they have ever been,” he claimed.

It was GitHub Copilot that made Welsh fundamentally reassess the meaning of computer science. “Copilot radically changes the way that we write code,” he said, “and has, at least in my personal experience, been just a remarkable and profound way of accelerating my development.”



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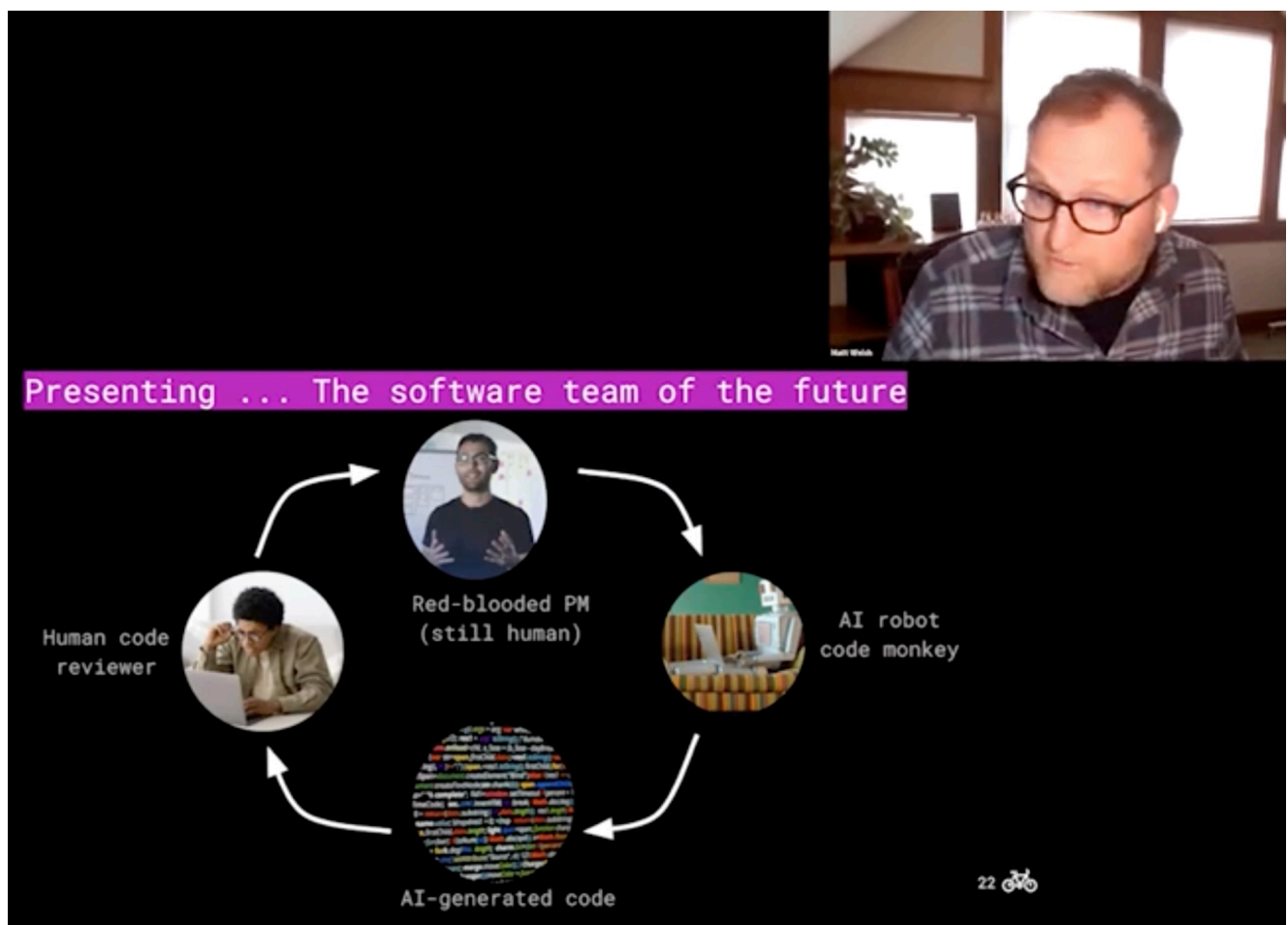
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more than I really think it should," he said. He added that Copilot is "a fantastic productivity boost because it saves me from having to context-switch."

As good as it is, Welsh thinks the current version of Copilot is "just the beginning." There are only two things stopping Copilot from getting much better, he said: more data and more compute. Since both of those things are abundant, he doesn't see "any reason why Copilot in a year or two, or maybe three, isn't going to get to the point of, you type a few lines at the top of a source file and it just writes the rest."

The Software Team of the Future

Welsh had some intriguing predictions about how this will change the team dynamics of software development. Basically, he only sees a couple of roles for humans once programmers are phased out: product manager and code reviewer.



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The role of product manager won't change much, he says.

Imagine a human product manager that is still able to write English-language descriptions — PRDs [product requirements documents], if you will — of what the

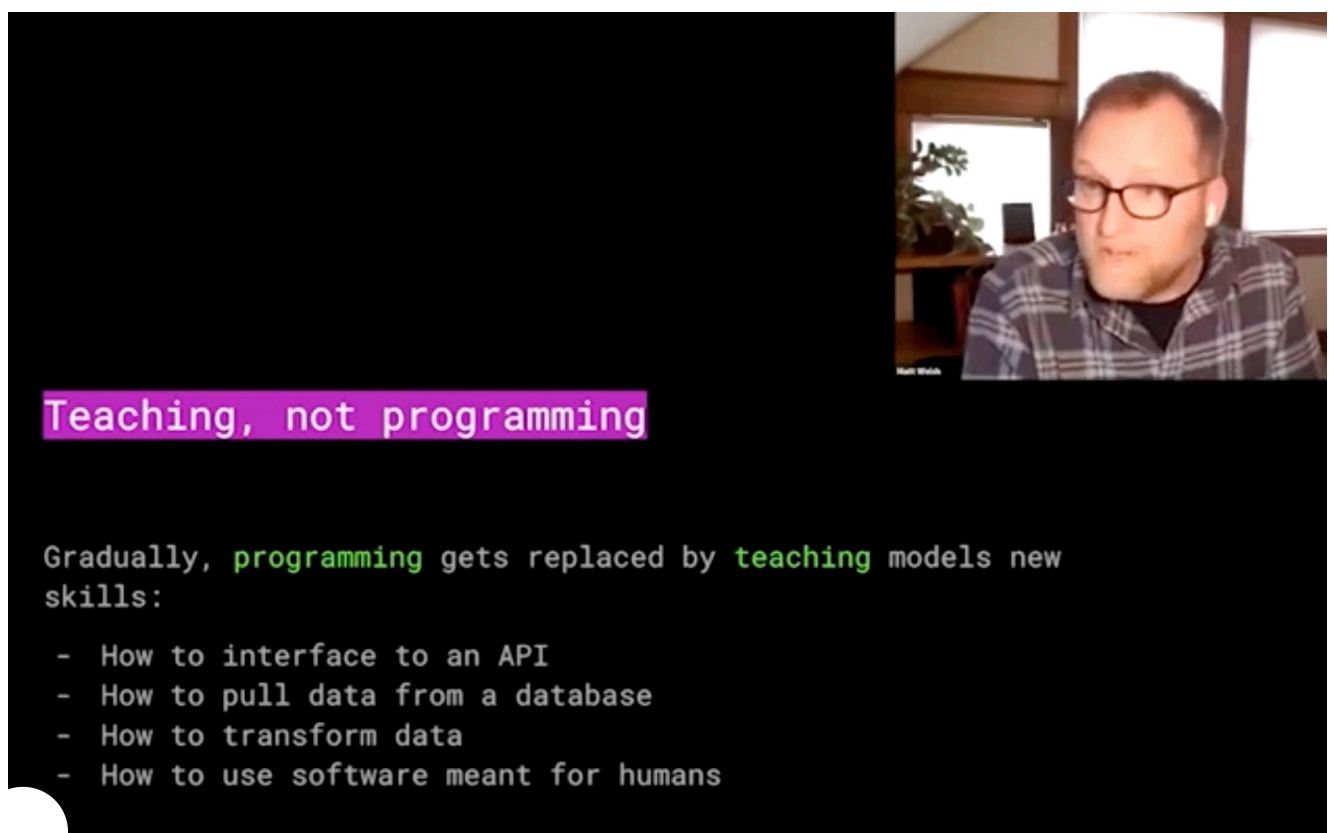
But in the near future, instead of handing a PRD to the engineering team and waiting six weeks or so for them to implement it, said Welsh, “you just hand the PRD to the AI, the AI spits out the code in, like, a few seconds.”

Of course, one of the early lessons of generative AI is that the output of these AIs won't necessarily be correct. So Welsh sees a role for human code reviewers in this programmer-less future.

“How do we know that the code works? How do we know that it's good? How do we know that it's right? Of course we need to have thorough testing, and the testing is very, very important — and that's not going away. All the CI/CD stuff and everything that we've invested in over the last few decades is still relevant here.”

He thinks that humans with programming ability will be tasked with “reviewing and reading the AI-generated code and making sure that it works, and is doing the right thing.”

As for programmers, and those about to join the field, they will need to become teachers of AI rather than coders. “It's all about teaching the AI, not writing the computer program,” said Welsh.



The image shows a video call in progress. On the right side, there is a small video window of a man with glasses and a beard, wearing a plaid shirt, identified as Matt Welsh. On the left side, there is a large black presentation slide with white and green text. The slide has a title 'Teaching, not programming' in a pink box, followed by a paragraph and a bulleted list.

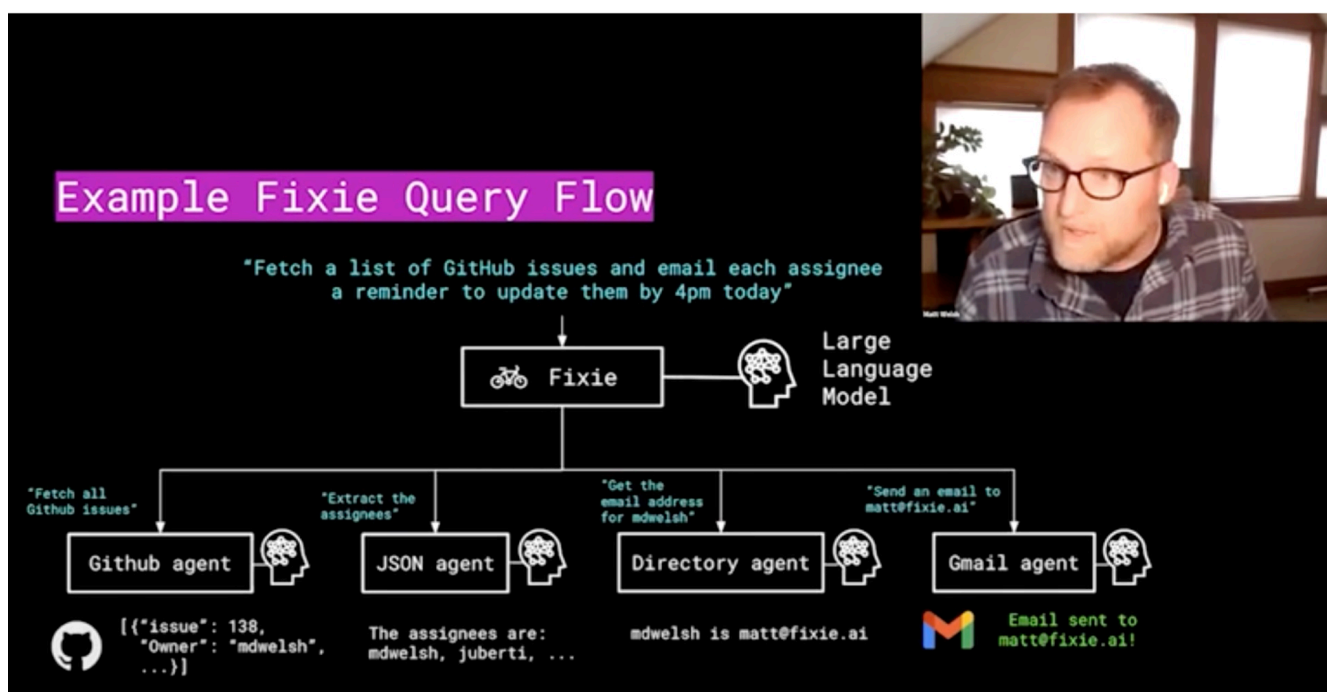
Teaching, not programming

Gradually, **programming** gets replaced by **teaching** models new skills:

- How to interface to an API
- How to pull data from a database
- How to transform data
- How to use software meant for humans

who tuned into this ACM meeting, Welsh then explained how his new company is taking advantage of the AI programming trend.

“With Fixie, the idea is you give it a description of what you want to do and Fixie takes it and, using a set of **large language models** *plus* agents that can connect to external systems, it can produce a result for you — either an answer to a question, or calling an API, or invoking a tool, or making a change in a database. These are all the things that these models can do. And critically, we’re doing this not by writing a whole ton of code; we’re doing this by teaching the AI models how to do this.”

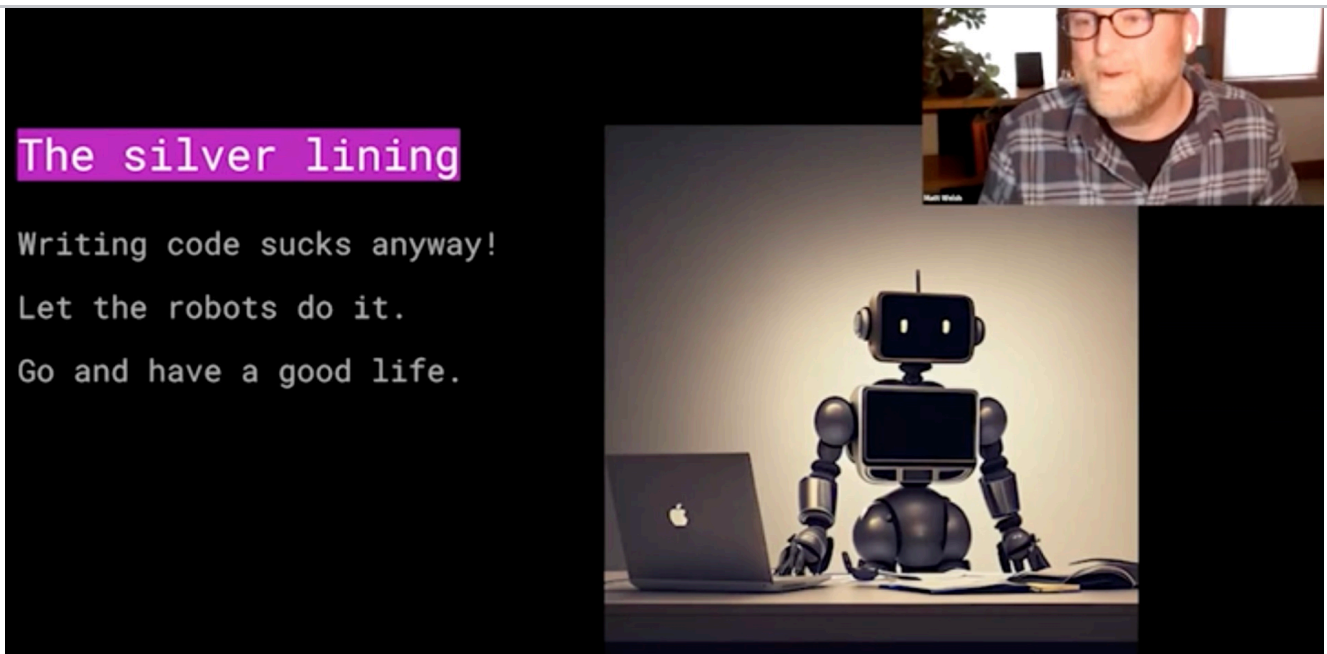


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Coding Sucks

It became clear in this presentation that Matt Welsh has a big vested interest in making traditional coding obsolete, since his company aims to capitalize on the trend of AI programming. However, given his own long and successful career as a computer scientist, you also have to take his predictions seriously. This is someone who has a deep understanding of how programming works, and he thinks AI is already close to mastering it.

Also, Welsh thinks that “coding sucks” anyway — so just let the robots do it. “Do something else with your time,” he suggests. “Writing computer programs is not the best use of time for everyone.”



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Easier said than done for people employed as programmers currently. But Matt Welsh is probably right that this is where coding is headed, so it's up to us humans to adapt.

Here's the full video of Matt Welsh's presentation for the Chicago ACM:

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Richard MacManus is a Senior Editor at The New Stack and writes about web and application development trends. Previously he founded ReadWriteWeb in 2003 and built it into one of the world's most influential technology news sites. From the early...

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